

Research Summary

My research focuses on **auditing the security and privacy of machine learning (ML) systems**, particularly large language models (LLMs). My work makes AI risks measurable across the AI lifecycle by (i) auditing **training-data leakage** through membership inference; (ii) tracing **data and model provenance** across ML development pipelines; and (iii) identifying and mitigating privacy risks arising from **LLM misuse**. I also develop data synthesis algorithms to protect sensitive information based on **differential privacy**.

Education

- 2023 – 2027 **Purdue University, USA**
(Expected) Ph.D. in Computer Science (Advisor: Ninghui Li)
- 2020 – 2023 **Zhejiang University, China**
M.E. in Computer Technology. *Rank 1/49.*
- 2016 – 2020 **East China Normal University, China**
B.E. in Data Science. *GPA 3.61/4 (rank 1/23)*

Selected Honors and Awards

- 2026 **National Winner**, Innovation Bowl (Purdue Team Lead)
- 2026 **NDSS Fellowship**, Internet Society (1 of 24 worldwide)
- 2023 **Ross Fellowship**, Purdue University
- 2023 **Herbold Scholarship**, Purdue University (1 of 7)
- 2023 **Presidential Doctoral Excellence Award**, Purdue University (1 of 150)
- 2023 **Excellent Master's Dissertation**, China (1 of 43)
- 2023 **Provincial Outstanding Graduate**, Zhejiang, China
- 2022 **National Scholarship**, China (0.1%)
- 2021 **National Scholarship**, China (0.1%)

Work Experience

- 2026.05 – **Research Intern, Perplexity AI**, Mentor: Dr. Ninghui Li
- Present Characterize privacy and safety risks of agents in real-world settings, design practical security guardrails for LLM agents, and establish a defense-in-depth architecture deployed in Perplexity's products.

- 2024.05 – 08 **Research Intern, Alibaba Qwen**, Mentor: Dr. Bolin Ding
Identified a new privacy threat to online pseudonymity arising from LLM misuse, and developed the *first* agentic framework demonstrating the feasibility of automated deanonymization with LLMs. Results published in ACL 2026 and the IEEE Data Engineering Bulletin.
- 2022 – 2023 **Research Intern, Microsoft Research Asia**, Mentor: Dr. Xing Xie
Designed a new explainable recommendation algorithm that combines the interpretability of sparse methods with the expressiveness of dense approaches. The algorithm was deployed on Xbox Gaming, with results published at WWW 2023. The project also resulted in a U.S. patent and received the Excellent Internship Award (top 10%).

Publications

First/Co-First **11** publications at CCS ($\times 2$), USENIX Security ($\times 2$), NDSS, ACL, VLDB, WWW, SIGIR, TKDE, and IEEE Data Eng. Bulletin.

Mentoring Students I mentored published **4** papers at USENIX Security, SIGIR, and ICDE ($\times 2$).

*equal contribution, †mentored student

- CCS 2026 **Yuntao Du**^{*}, Tanishq Praveen Pauskar^{†*}, Hao Wang, Jing Su, Ninghui Li. Privacy Leakage from a Thousand Words: Sub-Pixel Location Recovery from Dot Maps. In ACM Conference on Computer and Communications Security (CCS), 2026.
- ACL 2026 **Yuntao Du**, Zitao Li, Bolin Ding, Yaliang Li, Hanshen Xiao, Jingren Zhou, Ninghui Li. Automated Profile Inference with Language Model Agents. In Findings of the Association for Computational Linguistics (ACL), 2026.
- USENIX 2026 **Yuntao Du**, Yuetian Chen, Hanshen Xiao, Bruno Ribeiro, Ninghui Li. Imitative Membership Inference Attack. In USENIX Security Symposium (USENIX Security), 2026.
- USENIX 2026 Meng Tong^{†*}, **Yuntao Du**^{*}, Kejiang Chen, Weiming Zhang, Ninghui Li. Membership Inference Attacks on Tokenizers of Large Language Models. In USENIX Security Symposium (USENIX Security), 2026.
- USENIX 2026 Yuetian Chen, **Yuntao Du**, Kaiyuan Zhang, Ashish Kundu, Charles Fleming, Bruno Ribeiro, Ninghui Li. Window-based Membership Inference Attacks Against Fine-tuned Large Language Models. In USENIX Security Symposium (USENIX Security), 2026.
- ICLR 2026 Yuetian Chen, Kaiyuan Zhang, **Yuntao Du**, Edoardo Stoppa, Charles Fleming, Ashish Kundu, Bruno Ribeiro, Ninghui Li. Membership Inference Attacks Against Fine-tuned Diffusion Language Models. In International Conference on Learning Representations (ICLR), 2026.
- NDSS 2026 **Yuntao Du**, Jiacheng Li, Yuetian Chen, Kaiyuan Zhang, Zhizhen Yuan, Hanshen Xiao, Bruno Ribeiro, Ninghui Li. Cascading and Proxy Membership Inference Attacks. In Network and Distributed System Security Symposium (NDSS), 2026.
- IEEE Data Eng. Bulletin **Yuntao Du**, Zitao Li, Ninghui Li, Bolin Ding. Beyond Data Privacy: New Privacy Risks for Large Language Models. In IEEE Computer Society Data Engineering Bulletin. 2025.
- CCS 2025 **Yuntao Du**, Ninghui Li. Systematic Assessment of Tabular Data Synthesis. In ACM Conference on Computer and Communications Security (CCS), 2025.

- USENIX 2025 Kaiyuan Zhang, Siyuan Cheng, Hanxi Guo, Yuetian Chen, Zian Su, Shengwei An, **Yuntao Du**, Charles Fleming, Ashish Kundu, Xiangyu Zhang, Ninghui Li. SOFT: Selective Data Obfuscation for Protecting LLM Fine-tuning against Membership Inference Attacks. In USENIX Security Symposium (USENIX Security), 2025.

Before Ph.D.

- ICDE 2024 Yujia Hu[†], **Yuntao Du**, Zhikun Zhang, Ziquan Fang, Lu Chen, Kai Zheng, Yunjun Gao. Real-Time Trajectory Synthesis with Local Differential Privacy. In IEEE International Conference on Data Engineering (ICDE), 2024.
- VLDB 2023 **Yuntao Du**, Yujia Hu[†], Zhikun Zhang, Ziquan Fang, Lu Chen, Baihua Zheng, Yunjun Gao. LDPTTrace: Locally Differentially Private Trajectory Synthesis. In International Conference on Very Large Data Bases (VLDB), 2023.
- WWW 2023 **Yuntao Du**, Jianxun Lian, Jing Yao, Xiting Wang, Mingqi Wu, Lu Chen, Yunjun Gao, Xing Xie. Towards Explainable Collaborative Filtering with Taste Clusters Learning. In ACM Web Conference (WWW), 2023.
- SIGIR 2023 Xinjun Zhu[†], **Yuntao Du**, Yuren Mao, Lu Chen, Yujia Hu, Yunjun Gao. Knowledge-refined Denoising Network for Robust Recommendation. In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2023.
- ICDE 2023 Zhihao Zeng[†], **Yuntao Du**, Ziquan Fang, Lu Chen, Shiliang Pu, Guodong Chen, Hui Wang, Yunjun Gao. FLBooster: A Unified and Efficient Platform for Federated Learning Acceleration. In IEEE International Conference on Data Engineering (ICDE), 2023.
- TKDE 2023 Minjun Zhao, Lu Chen, Keyu Yang, **Yuntao Du**, Yunjun Gao. Finding Materialized Models for Model Reuse. In IEEE Transactions on Knowledge and Data Engineering (TKDE), 2023.
- TKDE 2023 **Yuntao Du**, Xinjun Zhu[†], Lu Chen, Ziquan Fang, Yunjun Gao. MetaKG: Meta-learning on Knowledge Graph for Cold-start Recommendation. In IEEE Transactions on Knowledge and Data Engineering (TKDE), 2023.
- KDD 2022 Ziquan Fang, **Yuntao Du**, Xinjun Zhu, Danlei Hu, Lu Chen, Yunjun Gao, Christian S. Jensen. Spatio-Temporal Trajectory Similarity Learning in Road Networks. In ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2022.
- SIGIR 2022 Yunjun Gao, **Yuntao Du**, Yujia Hu[†], Lu Chen, Xinjun Zhu, Baihua Zheng. Self-Guided Learning to Denoise for Robust Recommendation. In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2022.
- SIGIR 2022 **Yuntao Du**, Xinjun Zhu[†], Lu Chen, Baihua Zheng, Yunjun Gao. HAKG: Hierarchy-Aware Knowledge Gated Network for Recommendation. In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2022.
- ICDE 2021 Ziquan Fang, **Yuntao Du**, Lu Chen, Yujia Hu, Yunjun Gao, Gang Chen. E²DTC: An End to End Deep Trajectory Clustering Framework via Self-Training. In IEEE International Conference on Data Engineering (ICDE), 2021.
- VLDB 2021 Ziquan Fang, Lu Pan, Lu Chen, **Yuntao Du**, Yunjun Gao. MDTP: A Multi-source Deep Traffic Prediction Framework over Spatio-Temporal Trajectory Data. In International Conference on Very Large Data Bases (VLDB), 2021.

Patents

U.S. Patent Mingqi Wu, Jianxun Lian, **Yuntao Du**, Jing Yao, Xiting Wang, Bei Lu, Xing Xie.
Explainable Cluster-based Collaborative Filtering. U.S. Patent No. 12,524,502 B2, 2026.

Invited Talks

- 2026 Apr **CS 4501, Data Privacy**, *Guest lecture*, University of Virginia
Membership Inference Attacks: Progress and Directions
- 2026 Mar **CS 593, Trustworthy Machine Learning**, *Guest lecture*, Purdue University
Membership Inference Attacks: History, Recent Progress, and Future Directions
- 2026 Feb **NDSS 2026**
Cascading and Proxy Membership Inference Attacks
- 2025 Oct **ACM CCS 2025**
Systematic Assessment of Tabular Data Synthesis
- 2025 Jun **Research Trend**
Automated Profile Inference with Language Model Agents
- 2025 Mar **Graduate Research Symposium**, Purdue University
Automated Profile Inference with Language Model Agents
- 2025 Mar **CS 556, Data Security and Privacy**, *Guest lecture*, Purdue University
Differentially Private Data Synthesis
- 2023 Nov **AI Time**
Towards Explainable Collaborative Filtering with Taste Clusters Learning
- 2023 Jan **Microsoft Research Asia**
Explainable Clustering and Cluster-based Collaborative Filtering

Teaching & Mentoring

Teaching

Instructor, CS 556: Data Security and Privacy, Purdue University. Fall 2026
Teaching Assistant, CS 556: Data Security and Privacy, Purdue University. Spring 2025

Mentoring

Edward Chen, high school student, Carmel High School. 2026 – Present

Kimaya Deshpande, B.S. Computer Science, Purdue University. 2026 – Present

Annabelle Nakamura DeFosse, B.S. Computer Engineering, Purdue University. 2026 – Present

Patrick Li, B.S. Computer Science, Stanford University. 2025 – Present
Publications: 1 manuscript under submission

Tanishq Praveen Pauskar, B.S. Computer Science, Purdue University. 2025 – Present
Publications: CCS 2026

Meng Tong, M.S. Computer Science, University of Science and Technology of China. 2025 – Present
Publications: USENIX Security 2026

Hillary Yang, high school student, Carmel High School. 2024 – 2025
Publications: 1 manuscript at arXiv

Yujia Hu, M.S. Computer Science, Zhejiang University. 2022 – 2024
Publications: SIGIR 2022, VLDB 2023, ICDE 2024

Zhihao Zeng, M.S. Computer Science, Zhejiang University. 2022 – 2023
Publications: ICDE 2023

Xinjun Zhu, M.S. Computer Science, Zhejiang University. 2022 – 2023
Publications: SIGIR 2022, SIGIR 2023, TKDE 2023

Academic Service

Conference Reviewing

PC, USENIX Security Symposium	2027
PC, International Conference on Very Large Data Bases (VLDB)	2027
PC, ACM Asia Conference on Computer and Communications Security (AsiaCCS)	2027
PC, Neural Information Processing Systems (NeurIPS)	2026
Poster Committee, IEEE Symposium on Security and Privacy (S&P)	2026
PC, International Conference on Learning Representations (ICLR)	2025 – 2026
PC, International Conference on Artificial Intelligence and Statistics (AISTATS)	2025 – 2026
PC, ACM Web Conference (WWW)	2026
PC, ACM Conference on Data and Application Security and Privacy (CODASPY)	2026
PC, International Conference on Web Search and Data Mining (WSDM)	2026
PC, ACM Conference on Information and Knowledge Management (CIKM)	2024 – 2026
PC, ACM Conference on Research and Development in Information Retrieval (SIGIR)	2023 – 2026
PC, International Joint Conference on Neural Networks (IJCNN)	2026 – 2027
PC, Table Representation Learning Workshop (TRL)	2024 – 2025
PC, AAAI Conference on Artificial Intelligence (AAAI)	2023 – 2027
PC, ACM Conference on Information Retrieval in the Asia Pacific (SIGIR-AP)	2023 – 2024

Journal Reviewing

Reviewer, ACM Computing Surveys (CSUR)
Reviewer, IEEE Transactions on Dependable and Secure Computing (TDSC)
Reviewer, ACM Transactions on Privacy and Security (TOPS)
Reviewer, International Journal on Very Large Data Bases (VLDBJ)
Reviewer, ACM Transactions on Recommender Systems (TORS)
Reviewer, IEEE Transactions on Knowledge and Data Engineering (TKDE)
Reviewer, IEEE Transactions on Big Data (TBD)